Appendix: Theoretical Analysis of the Augmented Synthetic Historical Control Estimator (ASHC)

1 Setup and Assumptions

For the purposes of self-containment, let $\mathcal{T} = \{1, 2, \ldots, T\}$ index time at a monthly frequency. Define the pre-treatment period as $\mathcal{T}_1 := \{t \in \mathcal{T} : t \leq T_0\}$ and the post-treatment period as $\mathcal{T}_2 := \{t \in \mathcal{T} : t > T_0\}$. The number of post-treatment periods is $n = T - T_0$. Let $m \in \mathbb{N}$ denote the evaluation window length. Define the evaluation period as the final m months of the pre-treatment period $\mathcal{T}_{\text{eval}} := \{T_0 - m + 1, \ldots, T_0\} \subset \mathcal{T}_1$.

Let $\mathbf{y} = (y_1, y_2, \ldots, y_T)^\top \in \mathbb{R}^T$ denote the observed outcome vector for the treated unit. Define the pre-treatment outcome vector $\mathbf{y}_{\text{pre}} := \mathbf{y}_{\mathcal{T}_1} \in \mathbb{R}^{T_0}$, the evaluation subvector $\mathbf{y}_{\text{eval}} := \mathbf{y}_{\mathcal{T}_{\text{eval}}} \in \mathbb{R}^m$, and the post-treatment outcome vector $\mathbf{y}_{\text{post}} := \mathbf{y}_{\mathcal{T}_2} \in \mathbb{R}^n$. The evaluation subvector $\mathbf{y}_{\text{eval}}$ is the target pattern to be reconstructed using convex combinations of earlier segments from the pre-treatment period. SHC by Chen et al. [2024] constructs $N = T_0 - m - n + 1$ overlapping temporal segments of length m. Each segment $\widetilde{\mathbf{y}}_{[i,i+m-1]}$ is a contiguous subvector of the smoothed pre-treatment series, forming the donor matrix

$$\widetilde{\mathbf{Y}}_{\text{pre}} := \begin{bmatrix} \widetilde{\mathbf{y}}_{[1,m]} & \widetilde{\mathbf{y}}_{[2,m+1]} & \cdots & \widetilde{\mathbf{y}}_{[N,N+m-1]} \end{bmatrix} \in \mathbb{R}^{m \times N}.$$

Example 1.1. *Suppose $T_0 = 60$ is months of pre-treatment data and we want the segment length to be $m = 12$ months to predict a post-treatment horizon of $n = 6$ months. The number of eligible segments is:$N = 60 - 12 - 6 + 1 = 43$, where each segment is 12 months long.*

Respectively, SHC and ASHC solve:

$$\mathbf{w}^{\text{SHC}} = \underset{\mathbf{w} \in \mathbb{R}^N}{\operatorname{argmin}} \quad \left\| \widetilde{\mathbf{y}}_{\text{eval}} - \widetilde{\mathbf{Y}}_{\text{pre}} \mathbf{w} \right\|_2^2 + \varsigma \left\| \mathbf{C}_0^\top \mathbf{w} \right\|_2^2 \quad \text{subject to } \mathbf{w} \geq \mathbf{0}, \quad \mathbf{1}^\top \mathbf{w} = 1.$$

$$\mathbf{w}^{\text{ASHC}} = \arg \min_{\mathbf{w} \in \mathbb{R}^N} \left\{ \frac{1}{2\lambda} \|\widetilde{\mathbf{y}} - \widetilde{\mathbf{Y}} \mathbf{w}\|^2 + \|\mathbf{w} - \mathbf{w}^{\text{SHC}}\|^2 + \varsigma \|\mathbf{C}_0^\top \mathbf{w}\|_2^2 \right\} \quad \text{s.t. } \sum_{i=1}^N w_i = 1.$$

Here, $\lambda > 0$ is the regularization parameter, $\varsigma > 0$ is a penalty parameter, and $\mathbf{C}_0 \in \mathbb{R}^{N \times k}$ contains the eigenvectors of $\mathbf{G} = \widetilde{\mathbf{Y}}^\top \widetilde{\mathbf{Y}}$ corresponding to eigenvalues less than a threshold $\tau > 0$.

1.1 Assumptions

Assumption 1.1 (Linearity with noise)**.** *There exists $\mathbf{w}^* \in \mathbb{R}^N$ such that $\widetilde{\mathbf{y}} = \widetilde{\mathbf{Y}} \mathbf{w}^* + \boldsymbol{\eta}$, where $\boldsymbol{\eta} \in \mathbb{R}^m$ is a mean-zero noise vector.*

Assumption 1.2 (Bounded noise)**.** $\|\boldsymbol{\eta}\| \leq \delta$ *for some* $\delta > 0$.

Assumption 1.3 (Smoothness of latent trend, from Chen et al. [2024])**.** *Let $\ell(t)$ denote the latent smooth component of the treated potential outcome. Suppose $\ell(t)$ is $H + 1$ times differentiable with $|\ell^{(H+1)}(t)| < \epsilon$ for all t and some $H \geq 3$.*

Assumption 1.4 (Operator norm bound)**.** $\|\widetilde{\mathbf{Y}}\| \leq \gamma$.

Assumption 1.5 (Simplex feasibility)**.** $\mathbf{w}^{SHC} \in \Delta^{N-1} := \{\mathbf{w} \in \mathbb{R}^N : \sum w_i = 1,\ w_i \geq 0\}$.

Lemma 1.1 (Deviation from SHC)**.** *Let* $\Delta := \mathbf{w}^{ASHC} - \mathbf{w}^{SHC}$*. Then*

$$\|\Delta\| \ \leq \ \frac{\gamma}{2\lambda + 2\lambda\varsigma\|\mathbf{C}_0\|^2}\, \epsilon_{SHC}.$$

where $\epsilon_{SHC} := \|\widetilde{\mathbf{y}} - \widetilde{\mathbf{Y}}\mathbf{w}^{SHC}\|$ *and* $\|\mathbf{C}_0\|^2$ *is the operator norm of* $\mathbf{C}_0^\top \mathbf{C}_0$*.*

Proof. To analyze the ASHC optimization problem, recall the objective function:

$$J(\mathbf{w}) = \frac{1}{2\lambda}\|\widetilde{\mathbf{y}} - \widetilde{\mathbf{Y}}\mathbf{w}\|^2 + \|\mathbf{w} - \mathbf{w}^{\text{SHC}}\|^2 + \varsigma\|\mathbf{C}_0^\top \mathbf{w}\|^2, \tag{1.1}$$

subject to the affine constraint $\mathbf{1}^\top \mathbf{w} = 1$.

To enforce this constraint, we form th Lagrangian:

$$\mathcal{L}(\mathbf{w}, \mu) = J(\mathbf{w}) + \mu(\mathbf{1}^\top \mathbf{w} - 1), \tag{1.2}$$

where μ is a Lagrange multiplier ensuring feasibility of $\mathbf{w}$. We compute the gradient of $\mathcal{L}$ with respect to $\mathbf{w}$ by differentiating each term separately.

For the first term, define the residual vector:

$$r(\mathbf{w}) = \widetilde{\mathbf{y}} - \widetilde{\mathbf{Y}}\mathbf{w}. \tag{1.3}$$

The fit term is then:

$$\frac{1}{2\lambda}\|r(\mathbf{w})\|^2 = \frac{1}{2\lambda} r(\mathbf{w})^\top r(\mathbf{w}). \tag{1.4}$$

We can see that this is a standard quadratic form. We begin by applying the Chain Rule for vector calculus. The chain rule for a composite function $f(g(\mathbf{w}))$ is given by:

$$\nabla_{\mathbf{w}} f(g(\mathbf{w})) = \left(\nabla_{g(\mathbf{w})} f\right)^\top \nabla_{\mathbf{w}} g(\mathbf{w})$$

In our case, the inner function is the residual vector $g(\mathbf{w}) = r(\mathbf{w})$, and the outer function is the squared norm $f(r) = \frac{1}{2\lambda} r^\top r$. First, let's find the gradient of the outer function with respect to the residual vector r.

$$\nabla_r \left(\frac{1}{2\lambda} r^\top r\right) = \frac{1}{2\lambda}\nabla_r \left(\sum_i r_i^2\right) = \frac{1}{2\lambda}\begin{pmatrix} 2r_1 \\ 2r_2 \\ \vdots \end{pmatrix} = \frac{1}{\lambda} r$$

This is a well-known result: the gradient of $\|r\|^2$ is $2r$. The $\frac{1}{2\lambda}$ is a constant scaling factor.

Next, we need to find the gradient of the inner function, $r(\mathbf{w})$, with respect to the weight vector $\mathbf{w}$. This is the Jacobian matrix of the transformation.

$$\nabla_{\mathbf{w}} r(\mathbf{w}) = \nabla_{\mathbf{w}}(\widetilde{\mathbf{y}} - \widetilde{\mathbf{Y}}\mathbf{w})$$

Since $\widetilde{\mathbf{y}}$ is a constant vector with respect to $\mathbf{w}$, its gradient is zero. We only need to compute the gradient of $-\widetilde{\mathbf{Y}}\mathbf{w}$.

$$\nabla_{\mathbf{w}}(-\widetilde{\mathbf{Y}}\mathbf{w}) = -\widetilde{\mathbf{Y}}$$

This result comes from the general rule that for a linear function $A\mathbf{x}$, the gradient with respect to $\mathbf{x}$ is just the matrix A. Here, our linear function is a negative, so the gradient is $-\widetilde{\mathbf{Y}}$.

Now we plug the results from the above back into the chain rule formula.

$$\nabla_{\mathbf{w}} J(\mathbf{w}) = \left(\nabla_{r(\mathbf{w})} \frac{1}{2\lambda} r(\mathbf{w})^\top r(\mathbf{w})\right)^\top \nabla_{\mathbf{w}} r(\mathbf{w})$$

$$\nabla_{\mathbf{w}} J(\mathbf{w}) = \left(\frac{1}{\lambda} r(\mathbf{w})\right)^\top (-\widetilde{\mathbf{Y}})$$

Using the property that $(A\mathbf{x})^\top = \mathbf{x}^\top A^\top$, we get:

$$\nabla_{\mathbf{w}} J(\mathbf{w}) = \frac{1}{\lambda} r(\mathbf{w})^\top (-\widetilde{\mathbf{Y}}) = -\frac{1}{\lambda} r(\mathbf{w})^\top \widetilde{\mathbf{Y}}$$

This is equivalent to:

$$\nabla_{\mathbf{w}} J(\mathbf{w}) = -\frac{1}{\lambda} \widetilde{\mathbf{Y}}^\top r(\mathbf{w})$$

Finally, we substitute the definition of the residual vector, $r(\mathbf{w}) = \widetilde{\mathbf{y}} - \widetilde{\mathbf{Y}}\mathbf{w}$, back into the gradient expression.

$$\nabla_{\mathbf{w}} J(\mathbf{w}) = -\frac{1}{\lambda} \widetilde{\mathbf{Y}}^\top (\widetilde{\mathbf{y}} - \widetilde{\mathbf{Y}}\mathbf{w})$$

Distributing the negative sign gives us the final form:

$$\nabla_{\mathbf{w}} J(\mathbf{w}) = \frac{1}{\lambda} \widetilde{\mathbf{Y}}^\top (\widetilde{\mathbf{Y}}\mathbf{w} - \widetilde{\mathbf{y}})$$

Next, consider the deviation penalty term:

$$\|\mathbf{w} - \mathbf{w}^{\mathrm{SHC}}\|^2 = (\mathbf{w} - \mathbf{w}^{\mathrm{SHC}})^\top (\mathbf{w} - \mathbf{w}^{\mathrm{SHC}}). \tag{1.5}$$

Its gradient with respect to $\mathbf{w}$ is:

$$\nabla_{\mathbf{w}} \|\mathbf{w} - \mathbf{w}^{\mathrm{SHC}}\|^2 = 2(\mathbf{w} - \mathbf{w}^{\mathrm{SHC}}). \tag{1.6}$$

Notice that the Hessian of this term is $2\mathbf{I}$, because differentiating again with respect to $\mathbf{w}$ gives:

$$\nabla^2_{\mathbf{w}} \|\mathbf{w} - \mathbf{w}^{\mathrm{SHC}}\|^2 = 2\mathbf{I}. \tag{1.7}$$

Here $\mathbf{I}$ is the identity matrix, representing curvature equally in all coordinate directions. Thus the $2\mathbf{I}$ in M directly reflects the Hessian of this quadratic deviation penalty.

For the Gram penalty term, rewrite it as:

$$\varsigma \|\mathbf{C}_0^\top \mathbf{w}\|^2 = \varsigma \mathbf{w}^\top \mathbf{C}_0 \mathbf{C}_0^\top \mathbf{w}. \tag{1.8}$$

Since $\mathbf{C}_0 \mathbf{C}_0^\top$ is symmetric, its gradient is:

$$\nabla_{\mathbf{w}} \varsigma \mathbf{w}^\top \mathbf{C}_0 \mathbf{C}_0^\top \mathbf{w} = 2\varsigma \mathbf{C}_0 \mathbf{C}_0^\top \mathbf{w}. \tag{1.9}$$

Finally, the affine constraint term differentiates as:

$$\nabla_{\mathbf{w}} \mu(\mathbf{1}^\top \mathbf{w} - 1) = \mu \mathbf{1}. \tag{1.10}$$

We now have the first-order condition from the gradient of the Lagrangian:

$$\frac{1}{\lambda} \widetilde{\mathbf{Y}}^\top (\widetilde{\mathbf{Y}}\mathbf{w} - \widetilde{\mathbf{y}}) + 2(\mathbf{w} - \mathbf{w}^{\mathrm{SHC}}) + 2\varsigma \mathbf{C}_0 \mathbf{C}_0^\top \mathbf{w} + \mu \mathbf{1} = \mathbf{0}. \tag{1.11}$$

Expanding the deviation penalty term gives

$$2(\mathbf{w} - \mathbf{w}^{\mathrm{SHC}}) = 2\mathbf{w} - 2\mathbf{w}^{\mathrm{SHC}}. \tag{1.12}$$

Substituting back, the condition becomes

$$\frac{1}{\lambda}\widetilde{\mathbf{Y}}^\top\widetilde{\mathbf{Y}}\mathbf{w} - \frac{1}{\lambda}\widetilde{\mathbf{Y}}^\top\widetilde{\mathbf{y}} + 2\mathbf{w} - 2\mathbf{w}^{\mathrm{SHC}} + 2\varsigma\mathbf{C}_0\mathbf{C}_0^\top\mathbf{w} + \mu\mathbf{1} = \mathbf{0}. \tag{1.13}$$

Now collect all the terms involving $\mathbf{w}$ on the left-hand side:

$$\left(\frac{1}{\lambda}\widetilde{\mathbf{Y}}^\top\widetilde{\mathbf{Y}} + 2\mathbf{I} + 2\varsigma\mathbf{C}_0\mathbf{C}_0^\top\right)\mathbf{w} \tag{1.14}$$

and move all the other terms to the right-hand side:

$$\frac{1}{\lambda}\widetilde{\mathbf{Y}}^\top\widetilde{\mathbf{y}} + 2\mathbf{w}^{\mathrm{SHC}} - \mu\mathbf{1}. \tag{1.15}$$

Thus we obtain the compact matrix form:

$$\left(\frac{1}{\lambda}\widetilde{\mathbf{Y}}^\top\widetilde{\mathbf{Y}} + 2\mathbf{I} + 2\varsigma\mathbf{C}_0\mathbf{C}_0^\top\right)\mathbf{w} = \frac{1}{\lambda}\widetilde{\mathbf{Y}}^\top\widetilde{\mathbf{y}} + 2\mathbf{w}^{\mathrm{SHC}} - \mu\mathbf{1}. \tag{1.16}$$

Here the $2\mathbf{I}$ arises directly from the quadratic penalty $\|\mathbf{w} - \mathbf{w}^{\mathrm{SHC}}\|^2$, whose gradient contributes $2\mathbf{w}$, corresponding in matrix form to $2\mathbf{I}\cdot\mathbf{w}$.

Next, define the matrix

$$M = \frac{1}{\lambda}\widetilde{\mathbf{Y}}^\top\widetilde{\mathbf{Y}} + 2\mathbf{I} + 2\varsigma\mathbf{C}_0\mathbf{C}_0^\top. \tag{1.17}$$

M corresponds to the Hessian of the objective function $J(\mathbf{w})$ plus the identity-weighted contribution of the deviation penalty.

The solution $\mathbf{w}^{\mathrm{ASHC}}$ satisfies the first-order condition:

$$M\mathbf{w}^{\mathrm{ASHC}} = \frac{1}{\lambda}\widetilde{\mathbf{Y}}^\top\widetilde{\mathbf{y}} + 2\mathbf{w}^{\mathrm{SHC}} - \mu\mathbf{1}. \tag{1.18}$$

Evaluating the same matrix M at $\mathbf{w}^{\mathrm{SHC}}$, which is not necessarily the ASHC solution, gives:

$$M\mathbf{w}^{\mathrm{SHC}} = \frac{1}{\lambda}\widetilde{\mathbf{Y}}^\top\widetilde{\mathbf{Y}}\mathbf{w}^{\mathrm{SHC}} + 2\mathbf{w}^{\mathrm{SHC}} + 2\varsigma\mathbf{C}_0\mathbf{C}_0^\top\mathbf{w}^{\mathrm{SHC}}. \tag{1.19}$$

Subtracting this equation from the previous one, we get

$$M\mathbf{w}^{\mathrm{ASHC}} - M\mathbf{w}^{\mathrm{SHC}} = \left(\frac{1}{\lambda}\widetilde{\mathbf{Y}}^\top\widetilde{\mathbf{y}} + 2\mathbf{w}^{\mathrm{SHC}} - \mu\mathbf{1}\right) - \left(\frac{1}{\lambda}\widetilde{\mathbf{Y}}^\top\widetilde{\mathbf{Y}}\mathbf{w}^{\mathrm{SHC}} + 2\mathbf{w}^{\mathrm{SHC}} + 2\varsigma\mathbf{C}_0\mathbf{C}_0^\top\mathbf{w}^{\mathrm{SHC}}\right). \tag{1.20}$$

Simplifying the right-hand side by canceling the $2\mathbf{w}^{\mathrm{SHC}}$ terms, we obtain

$$M(\mathbf{w}^{\mathrm{ASHC}} - \mathbf{w}^{\mathrm{SHC}}) = \frac{1}{\lambda}\widetilde{\mathbf{Y}}^\top\left(\widetilde{\mathbf{y}} - \widetilde{\mathbf{Y}}\mathbf{w}^{\mathrm{SHC}}\right) - \mu\mathbf{1}. \tag{1.21}$$

Define the deviation

$$\Delta = \mathbf{w}^{\mathrm{ASHC}} - \mathbf{w}^{\mathrm{SHC}}, \tag{1.22}$$

and the residual vector

$$r = \widetilde{\mathbf{y}} - \widetilde{\mathbf{Y}}\mathbf{w}^{\mathrm{SHC}}. \tag{1.23}$$

Since both $\mathbf{w}^{\mathrm{ASHC}}$ and $\mathbf{w}^{\mathrm{SHC}}$ satisfy the affine constraint $\mathbf{1}^\top\mathbf{w} = 1$, their difference Δ lies in the hyperplane orthogonal to $\mathbf{1}$:

$$\mathbf{1}^\top\Delta = 0. \tag{1.24}$$

Taking the inner product of both sides of the previous equation with Δ gives

$$\Delta^\top M\Delta = \frac{1}{\lambda}\Delta^\top \widetilde{\mathbf{Y}}^\top r - \mu \underbrace{\Delta^\top \mathbf{1}}_{=0} = \frac{1}{\lambda}\Delta^\top \widetilde{\mathbf{Y}}^\top r. \tag{1.25}$$

Here, the right-hand side is an inner product between the deviation vector Δ and the vector $\widetilde{\mathbf{Y}}^\top r$. Because inner products can be difficult to control directly, we apply the Cauchy–Schwarz inequality to bound it:

$$|\Delta^\top \widetilde{\mathbf{Y}}^\top r| \leq \|\Delta\| \, \|\widetilde{\mathbf{Y}}^\top r\|. \tag{1.26}$$

Thus, $\Delta^\top M\Delta \leq \frac{1}{\lambda}\|\Delta\| \, \|\widetilde{\mathbf{Y}}^\top r\|$. By assumption, the operator norm of $\widetilde{\mathbf{Y}}$ is bounded by γ, and the residual norm is $\epsilon_{\text{SHC}} = \|r\|$, so

$$\|\widetilde{\mathbf{Y}}^\top r\| \leq \|\widetilde{\mathbf{Y}}\| \|r\| \leq \gamma\epsilon_{\text{SHC}}. \tag{1.27}$$

Thus, $\Delta^\top M\Delta \leq \frac{\gamma}{\lambda}\|\Delta\|\epsilon_{\text{SHC}}$. On the other hand, since M is positive definite, its smallest eigenvalue $\lambda_{\min}(M)$ guarantees $\Delta^\top M\Delta \geq \lambda_{\min}(M)\|\Delta\|^2$.

Putting these two inequalities together, we sandwich the quadratic form:

$$\lambda_{\min}(M)\|\Delta\|^2 \leq \Delta^\top M\Delta \leq \frac{\gamma}{\lambda}\|\Delta\| \, \epsilon_{\text{SHC}}$$

The left-hand side is the lower bound derived from the smallest eigenvalue of the positive definite matrix M. The right-hand side is the upper bound derived from the Cauchy-Schwarz and operator norm inequalities. We can now discard the middle term to focus on the key inequality between the two bounds:

$$\lambda_{\min}(M)\|\Delta\|^2 \leq \frac{\gamma}{\lambda}\|\Delta\| \, \epsilon_{\text{SHC}}$$

Assuming $\|\Delta\| > 0$ (the trivial case where the weights are identical is already clear), we can divide both sides of the inequality by $\|\Delta\|$ to isolate the term of interest.

$$\lambda_{\min}(M)\|\Delta\| \leq \frac{\gamma}{\lambda}\epsilon_{\text{SHC}}$$

Rearranging this inequality to solve for $\|\Delta\|$ gives us an intermediate bound:

$$\|\Delta\| \leq \frac{\gamma}{\lambda\lambda_{\min}(M)}\epsilon_{\text{SHC}}$$

Next, we establish a lower bound for the smallest eigenvalue of the matrix M. Recall that the matrix M is defined as:

$$M = \frac{1}{\lambda}\widetilde{\mathbf{Y}}^\top \widetilde{\mathbf{Y}} + 2\mathbf{I} + 2\varsigma\mathbf{C}_0\mathbf{C}_0^\top$$

The smallest eigenvalue of a sum of positive semi-definite matrices is at least the sum of their smallest eigenvalues. Since the first term's smallest eigenvalue is non-negative, and the second term's is 2, we have:

$$\lambda_{\min}(M) \geq \lambda_{\min}\left(\frac{1}{\lambda}\widetilde{\mathbf{Y}}^\top \widetilde{\mathbf{Y}}\right) + \lambda_{\min}(2\mathbf{I}) + \lambda_{\min}(2\varsigma\mathbf{C}_0\mathbf{C}_0^\top)$$

$$\lambda_{\min}(M) \geq 0 + 2 + 2\varsigma\lambda_{\min}(\mathbf{C}_0\mathbf{C}_0^\top)$$

By the assumption that $\lambda_{\min}(\mathbf{C}_0\mathbf{C}_0^\top) \geq \|\mathbf{C}_0\|^2$, we get the lower bound for the smallest eigenvalue of M:

$$\lambda_{\min}(M) \geq 2 + 2\varsigma\|\mathbf{C}_0\|^2$$

Finally, we substitute this lower bound for $\lambda_{\min}(M)$ back into the intermediate bound for $\|\Delta\|$. A smaller value in the denominator leads to a larger (and therefore valid) upper bound on the expression. This gives us the final bound on the deviation:

$$\|\Delta\| \leq \frac{\gamma}{\lambda(2 + 2\varsigma\|\mathbf{C}_0\|^2)} \epsilon_{\text{SHC}} = \frac{\gamma}{2\lambda + 2\lambda\varsigma\|\mathbf{C}_0\|^2} \epsilon_{\text{SHC}}$$

This final expression shows that the deviation between the ASHC and SHC weights is controlled by the SHC residual error, scaled by a factor that depends on the regularization parameters λ and ς. ☐

Corollary 1.1. *The primary driver of the deviation $\|\Delta\|$ between the ASHC and SHC weights is the SHC residual error ϵ_{SHC}. The terms λ and $\lambda_{\min}(M)$ appear in the denominator and therefore act only as scaling factors: stronger regularization (λ large) or greater curvature of the Hessian ($\lambda_{\min}(M)$ large) shrink the deviation, but cannot generate deviation on their own. In contrast, ϵ_{SHC} enters multiplicatively in the numerator. Thus, if the SHC fit is already tight (so that ϵ_{SHC} is small), the deviation $\|\Delta\|$ is necessarily small. Conversely, if SHC exhibits poor fit, then ϵ_{SHC} is large, and ASHC permits correspondingly larger deviations, though still bounded by the regularization terms. In this way, ASHC improves stability without moving far from the baseline SHC solution unless the SHC fit leaves room for meaningful improvement.*

Lemma 1.2. *Under (A4), the error from deviating off SHC weights satisfies*

$$\|\widetilde{\mathbf{Y}}\Delta\| \leq \gamma\|\Delta\|.$$

Proof. Recall that $\Delta = \mathbf{w}^{\text{ASHC}} - \mathbf{w}^{\text{SHC}} \in \mathbb{R}^N$ denotes the deviation of ASHC weights from the SHC weights. To understand how much this weight deviation changes the fitted donor outcomes, we consider the vector $\widetilde{\mathbf{Y}}\Delta$.

From Assumption (A4), the donor matrix $\widetilde{\mathbf{Y}}$ has operator (spectral) norm bounded by γ:

$$\|\widetilde{\mathbf{Y}}\| \ \leq \ \gamma.$$

The operator norm of a matrix is defined as

$$\|\widetilde{\mathbf{Y}}\| \ = \ \sup_{\mathbf{v}\neq 0} \frac{\|\widetilde{\mathbf{Y}}\mathbf{v}\|}{\|\mathbf{v}\|}.$$

This measures the maximum stretching effect that the matrix $\widetilde{\mathbf{Y}}$ can have on any vector $\mathbf{v}$. In other words, it tells us how much $\widetilde{\mathbf{Y}}$ can amplify a deviation in the weights.

Applying this definition with $\mathbf{v} = \Delta$ gives

$$\frac{\|\widetilde{\mathbf{Y}}\Delta\|}{\|\Delta\|} \ \leq \ \|\widetilde{\mathbf{Y}}\| \ \leq \ \gamma.$$

Multiplying through by $\|\Delta\|$ yields

$$\|\widetilde{\mathbf{Y}}\Delta\| \ \leq \ \gamma\|\Delta\|.$$

Thus, even if the ASHC weights deviate from the SHC weights, the resulting change in fitted donor outcomes cannot exceed γ times the size of that deviation. This ensures that the donor matrix does not amplify weight differences in an uncontrolled way. ☐

Lemma 1.3. *Under (A3), for horizon $k > 0$,*

$$|\ell(T_0 + k) - \ell^{SHC}(T_0 + k)| \ \leq \ b_\epsilon(H, k) \ := \ \frac{2\epsilon|k|^{H+1}}{(H+1)!}.$$

Proof. We show that the SHC approximation error in predicting the smooth latent trend $\ell(\cdot)$ is of order $|k|^{H+1}$. The argument proceeds in four steps.

Step 1. Taylor expansion of the latent trend. Let $\ell(t)$ denote the latent smooth component of the treated outcome. By Assumption (A3), $\ell(t)$ is $(H+1)$ times differentiable with $|\ell^{(H+1)}(t)| < \epsilon$ for all t. Expanding ℓ around T_0 using Taylor's theorem gives

$$\ell(T_0 + k) = \sum_{h=0}^{H} \frac{\ell^{(h)}(T_0)}{h!}\, k^h \;+\; R_{H+1}(k),$$

where the remainder term is

$$R_{H+1}(k) = \frac{\ell^{(H+1)}(\xi)}{(H+1)!}\, k^{H+1}, \quad \text{for some } \xi \in [T_0, T_0 + k].$$

Step 2. Bound the Taylor remainder. Since $|\ell^{(H+1)}(t)| \leq \epsilon$ for all t, the remainder is bounded by

$$|R_{H+1}(k)| \;\leq\; \frac{\epsilon}{(H+1)!}\, |k|^{H+1}.$$

Step 3. Representation of the SHC approximation. The SHC predictor $\ell^{\mathrm{SHC}}(T_0 + k)$ is constructed as a convex combination of donor segments. By design, these donor segments come from smoothed pre-treatment trajectories that share the same latent trend structure $\ell(\cdot)$. Thus, when we form ℓ^{SHC}, the lower-order terms of the Taylor expansion (up to order H) are matched exactly in expectation, and the residual discrepancy is of the same order as the Taylor remainder.

Formally, we may write

$$\ell^{\mathrm{SHC}}(T_0 + k) = \sum_{h=0}^{H} \frac{\ell^{(h)}(T_0)}{h!}\, k^h \;+\; R'_{H+1}(k),$$

where $R'_{H+1}(k)$ is another error term of order $|k|^{H+1}$.

Step 4. Combine the two remainder terms. Subtracting the SHC expansion from the true expansion gives

$$\ell(T_0 + k) - \ell^{\mathrm{SHC}}(T_0 + k) = R_{H+1}(k) - R'_{H+1}(k).$$

Applying the triangle inequality,

$$|\ell(T_0 + k) - \ell^{\mathrm{SHC}}(T_0 + k)| \;\leq\; |R_{H+1}(k)| + |R'_{H+1}(k)|.$$

Each remainder is bounded by $\frac{\epsilon}{(H+1)!}\, |k|^{H+1}$, so together we obtain

$$|\ell(T_0 + k) - \ell^{\mathrm{SHC}}(T_0 + k)| \;\leq\; \frac{2\epsilon}{(H+1)!}\, |k|^{H+1}.$$

This completes the proof. □

Remark 1.1. *Lemma 1.3 formalizes the idea that the SHC predictor inherits the smoothness of the latent trend $\ell(\cdot)$. Because SHC is built from overlapping donor segments that reflect the same underlying smooth dynamics, the approximation reproduces the first H terms of the Taylor expansion around T_0. The only discrepancy comes from the $(H+1)$-st order remainder, which is small when the latent trend is sufficiently smooth. Thus, the difference between ℓ and ℓ^{SHC} grows only polynomially in the forecast horizon k, and the error is of order $|k|^{H+1}$ rather than linear or quadratic in k. This ensures that SHC extrapolates the smooth component accurately over short to moderate horizons.*

2 Main Result

Proposition 2.1. *Under assumptions (A1)–(A5), the ASHC prediction error is bounded by*

$$\|\widetilde{\mathbf{y}} - \widetilde{\mathbf{Y}}\mathbf{w}^{ASHC}\| \;\leq\; \epsilon_{SHC}\left(1 + \frac{\gamma}{2\lambda + 2\varsigma\|\mathbf{C}_0\|^2}\right) + b_\epsilon(H,k) + \delta.$$

Proof. We want to establish an upper bound on the prediction error of the ASHC estimator,

$$\|\widetilde{\mathbf{y}} - \widetilde{\mathbf{Y}}\mathbf{w}^{\mathrm{ASHC}}\|.$$

The argument proceeds in four steps.

Step 1. Break down the error into two components. Let $\mathbf{w}^{\mathrm{SHC}}$ denote the SHC weights and define

$$\Delta := \mathbf{w}^{\mathrm{ASHC}} - \mathbf{w}^{\mathrm{SHC}}.$$

By the triangle inequality,

$$\|\widetilde{\mathbf{y}} - \widetilde{\mathbf{Y}}\mathbf{w}^{\mathrm{ASHC}}\| \;\leq\; \underbrace{\|\widetilde{\mathbf{y}} - \widetilde{\mathbf{Y}}\mathbf{w}^{\mathrm{SHC}}\|}_{\epsilon_{\mathrm{SHC}}} + \|\widetilde{\mathbf{Y}}\Delta\|.$$

Step 2. Bound the contribution of $\|\widetilde{\mathbf{Y}}\Delta\|$. By Lemma 1.2, we know that

$$\|\widetilde{\mathbf{Y}}\Delta\| \;\leq\; \gamma\|\Delta\|.$$

Step 3. Bound the deviation $\|\Delta\|$. By Lemma 1.1, the deviation satisfies

$$\|\Delta\| \;\leq\; \frac{\epsilon_{\mathrm{SHC}}}{2\lambda + 2\varsigma\|\mathbf{C}_0\|^2}.$$

Step 4. Account for smoothness bias and noise. From Lemma 1.3, the smoothness bias is bounded by

$$b_\epsilon(H,k) \;=\; \frac{2\epsilon|k|^{H+1}}{(H+1)!}.$$

From Assumption (A2), the noise contributes at most

$$\delta.$$

Step 5. Putting the pieces together. Substituting the bound from Step 2 into Step 1 gives:

$$\|\widetilde{\mathbf{y}} - \widetilde{\mathbf{Y}}\mathbf{w}^{\mathrm{ASHC}}\| \;\leq\; \epsilon_{\mathrm{SHC}} + \gamma\|\Delta\|.$$

Now substitute the bound for $\|\Delta\|$ from Step 3:

$$\|\widetilde{\mathbf{y}} - \widetilde{\mathbf{Y}}\mathbf{w}^{\mathrm{ASHC}}\| \;\leq\; \epsilon_{\mathrm{SHC}} + \gamma\cdot\frac{\epsilon_{\mathrm{SHC}}}{2\lambda + 2\varsigma\|\mathbf{C}_0\|^2}.$$

Next, add the smoothness bias and noise bounds from Step 4:

$$\|\widetilde{\mathbf{y}} - \widetilde{\mathbf{Y}}\mathbf{w}^{\mathrm{ASHC}}\| \;\leq\; \epsilon_{\mathrm{SHC}} + \gamma\cdot\frac{\epsilon_{\mathrm{SHC}}}{2\lambda + 2\varsigma\|\mathbf{C}_0\|^2} + b_\epsilon(H,k) + \delta.$$

Step 6. Simplify the bound. Factor out ϵ_{SHC} from the first two terms:

$$\epsilon_{\mathrm{SHC}} + \gamma \cdot \frac{\epsilon_{\mathrm{SHC}}}{2\lambda + 2\varsigma\|\mathbf{C}_0\|^2} \;=\; \epsilon_{\mathrm{SHC}} \left(1 + \frac{\gamma}{2\lambda + 2\varsigma\|\mathbf{C}_0\|^2}\right).$$

Therefore,

$$\|\widetilde{\mathbf{y}} - \widetilde{\mathbf{Y}}\mathbf{w}^{\mathrm{ASHC}}\| \;\leq\; \epsilon_{\mathrm{SHC}} \left(1 + \frac{\gamma}{2\lambda + 2\varsigma\|\mathbf{C}_0\|^2}\right) + b_\epsilon(H,k) + \delta.$$

This completes the proof. □

Remark 2.2. *The proof explicitly shows how each component contributes to the final bound: (1) the SHC baseline error, (2) the ASHC deviation penalty controlled by λ and ς, (3) the smoothness bias from Taylor approximation, and (4) the bounded noise. Factoring ϵ_{SHC} at the end highlights how ASHC scales the SHC error by at most a small multiplicative factor depending on the regularization strength.*

3 Asymptotic Results

Theorem 3.1. *Under (A1)–(A5), as $m \to \infty$ and $H \to \infty$, for fixed $k > 0$,*

$$\|\widetilde{\mathbf{y}} - \widetilde{\mathbf{Y}}w^{ASHC}\| \longrightarrow \delta.$$

If $\boldsymbol{\eta}$ consists of independent mean-zero sub-Gaussian entries, then

$$\|\widetilde{\mathbf{y}} - \widetilde{\mathbf{Y}}w^{ASHC}\| \xrightarrow{p} 0.$$

Proof. Starting from Proposition 2.1, we have

$$\|\widetilde{\mathbf{y}} - \widetilde{\mathbf{Y}}w^{\mathrm{ASHC}}\| \leq \epsilon_{\mathrm{SHC}} \left(1 + \frac{\gamma}{\sqrt{2\lambda + 2\varsigma\|\mathbf{C}_0\|^2}}\right) + b_\epsilon(H,k) + \delta.$$

Step 1. Behavior of ϵ_{SHC}. By (A3), $\ell(t)$ is $(H+1)$-times differentiable. Gaussian kernel smoothing in SHC ensures

$$\epsilon_{\mathrm{SHC}} \to 0 \quad \text{as } m \to \infty,\ H \to \infty.$$

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Step 2. Behavior of ASHC penalty term. Since $\epsilon_{\mathrm{SHC}} \to 0$, the inflation factor

$$\frac{\gamma}{\sqrt{2\lambda + 2\varsigma\|\mathbf{C}_0\|^2}}\,\epsilon_{\mathrm{SHC}}$$

vanishes asymptotically.

Step 3. Behavior of smoothness bias $b_\epsilon(H,k)$. From Taylor's theorem,

$$\ell(T_0 + k) = \sum_{h=0}^{H} \frac{\ell^{(h)}(T_0)}{h!} k^h + R_{H+1}(k),$$

with

$$|R_{H+1}(k)| \leq \frac{\epsilon}{(H+1)!} |k|^{H+1}.$$

Thus

$$b_\epsilon(H,k) \leq \frac{2\epsilon}{(H+1)!} |k|^{H+1} \to 0 \quad \text{as } H \to \infty.$$

Step 4. Behavior of noise term. By (A2), $\|\boldsymbol{\eta}\| \le \delta$. If $\boldsymbol{\eta}$ is i.i.d. sub-Gaussian with variance proxy σ^2, then

$$\frac{\|\boldsymbol{\eta}\|}{m} \xrightarrow{p} 0.$$

Step 5. Combine bounds. Therefore,

$$\limsup_{m\to\infty} \|\widetilde{\mathbf{y}} - \widetilde{\mathbf{Y}} w^{\mathrm{ASHC}}\| \le \delta.$$

With sub-Gaussian noise, the limit is 0 in probability. □

Corollary 3.2. *Under (A1)–(A5), the ASHC estimator is asymptotically unbiased up to δ. If $\boldsymbol{\eta}$ is i.i.d. mean-zero sub-Gaussian with variance proxy σ^2 and bandwidth $h_m \asymp m^{-1/(2H+3)}$, then as $m \to \infty$,*

$$\|\widetilde{\mathbf{y}} - \widetilde{\mathbf{Y}} w^{ASHC}\| = O_p\Big(m^{-H/(2H+3)}\Big).$$

For large H, this approaches the parametric rate $O_p(m^{-1/2})$.

Proof. **Part 1: Consistency.** From Theorem 3.1, the error converges to δ, or 0 in probability under sub-Gaussian noise.

Part 2: Convergence Rate. From Proposition 2.1, we have

$$\|\widetilde{\mathbf{y}} - \widetilde{\mathbf{Y}} w^{\mathrm{ASHC}}\| \le \epsilon_{\mathrm{SHC}} \left(1 + \frac{\gamma}{\sqrt{2\lambda + 2\varsigma \|\mathbf{C}_0\|^2}}\right) + b_\epsilon(H,k) + \delta.$$

Step 1. Bias term. From Lemma 1.3,

$$b_\epsilon(H,k) = O(h_m^H).$$

Step 2. Stochastic error term. With sub-Gaussian noise, the smoothed variance is

$$O_p\big((mh_m)^{-1/2}\big).$$

Step 3. SHC fit error. Combining,

$$\epsilon_{\mathrm{SHC}} = O_p\big(h_m^H + (mh_m)^{-1/2}\big).$$

Step 4. Optimal bandwidth. Equating orders $h_m^H \asymp (mh_m)^{-1/2}$ yields

$$h_m \asymp m^{-1/(2H+3)}.$$

Step 5. Convergence rate. Substituting,

$$\epsilon_{\mathrm{SHC}} = O_p\big(m^{-H/(2H+3)}\big).$$

Thus the total error is

$$\|\widetilde{\mathbf{y}} - \widetilde{\mathbf{Y}} w^{\mathrm{ASHC}}\| = O_p\big(m^{-H/(2H+3)}\big).$$

Step 6. Near-parametric rate. As $H \to \infty$,

$$\frac{H}{2H+3} \to \tfrac{1}{2}.$$

So the rate approaches $O_p(m^{-1/2})$. □

Remark 3.3. *The deterministic noise bound δ in (A2) corresponds to $\|\boldsymbol{\eta}\| \le \delta$. Under sub-Gaussian noise, $\delta = O(\sigma\sqrt{m})$, while kernel smoothing reduces effective noise to $O_p((mh_m)^{-1/2})$. This explains the appearance of the nonparametric rate.*

References

Yi-Ting Chen, Jui-Chung Yang, and Tzu-Ting Yang. Synthetic historical control for policy evaluation. Available at SSRN: https://ssrn.com/abstract=4995085, September 2024. URL http://dx.doi.org/10.2139/ssrn.4995085.